Quantification: 542_z06
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PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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measure	value
lacuna count	19
interior lacuna count	15
frame-edge lacuna count	4

Interior lacunae only: mean, median and sample SD (n = 15)
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measure	kind	mean	median	sd	units
area_px2	lacuna shape	2576	3013	1380.42	px^2
major_axis_length_px	lacuna shape	109.567	124.464	38.2072	px
minor_axis_length_px	lacuna shape	31.8891	33.1799	10.1914	px
aspect_ratio	lacuna shape	3.6142	3.6285	1.2579	unitless
eccentricity	lacuna shape	0.9409	0.9613	0.0498	unitless
solidity	lacuna shape	0.7493	0.7672	0.1286	unitless
roots_count	headline	15.8	16	7.1933	count
ring_length_r30_px	headline	514.4	480	205.48	px
ring_length_r60_px	reported	1084.07	1012	453.743	px
owned_length_px	ownership-dependent	2256.74	2110.11	1399.82	px
edge_count	ownership-dependent	104.2	95	65.6715	count
mean_edge_length_px	ownership-dependent	21.5941	20.8459	2.8458	px
perimeter_px	normalised	292.07	309.522	122.808	px
ring_area_r30_px2	normalised	10491.5	11337	3304.03	px^2
ring_area_r60_px2	normalised	24400.9	26961	7815.16	px^2
in_frame_fraction_r30	normalised	1	1	0.0001	unitless
in_frame_fraction_r60	normalised	0.9858	1	0.0377	unitless
ring_density_r30	normalised	0.0482511	0.0479858	0.00612561	px^-1
ring_density_r60	normalised	0.0432981	0.042357	0.00658166	px^-1
roots_per_100px_perimeter	normalised	5.7574	5.5065	1.7381	per 100 px
ring_attached_length_r30_px	network v2	477.8	436	206.785	px
ring_attached_length_r60_px	network v2	931.2	846	466.423	px
ring_length_w_r30_px	network v2	585.895	539.375	237.469	px
ring_length_w_r60_px	network v2	1231.85	1135.48	521.155	px
sholl_crossings_r10	network v2	11.1333	11	5.2762	count
sholl_crossings_r20	network v2	13.9333	13	5.5609	count
sholl_crossings_r30	network v2	14	12	5.2915	count
loop_count	loops, report only	8.6667	7	7.8072	count
loop_edge_share	loops, report only	0.2629	0.2439	0.1041	share
ring_width_mean_r30_px	width	7.6115	7.6042	0.4582	px

Per field
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measure	value	units
analysed_area_px2	1.0064e+06	px^2
total_skeleton_length_px	41330	px
canalicular_length_density_per_px	0.0410671	px^-1
junction_count	972	count
junction_density_per_px2	0.00096582	px^-2
lacuna_count	19	count

lacunae_per_px2	1.888e-05	px^-2
skeleton_component_count	221	count
mean_component_length_px	187.014	px
median_component_length_px	25	px
field_length_density_w_per_px	0.0465273	px^-1
field_density_without_flagged_per_px	0.0409871	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	344	px
unattached_component_count	18	count
canalicular_length_density_attached_per_px	0.0410671	px^-1
field_length_density_with_unattached_per_px	0.0414089	px^-1
loop_count	167	count
loop_edge_share	0.3274	share
edge_count	2080	count

Canalicular width, the new measure

measure	value	units
width_mean_px	8.2041	px
width_median_px	8.2462	px
width_p10_px	4.4721	px
width_p90_px	10.7703	px
width_pixels_used	39992	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel. Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	1381	54.5303	43.8441	1.2437	0.5946	0.657	-0.6304	16.7227	909.274	True
2	445	56.9837	11.5357	4.9398	0.9793	0.7672	-0.2356	148.654	554.791	False
3	1001	62.1533	28.1242	2.21	0.8918	0.5533	-0.1968	55.7832	107.949	False
4	895	46.7108	27.3564	1.7075	0.8106	0.7956	0.2016	143.341	635.06	False
5	2450	125.59	27.3191	4.5971	0.9761	0.6618	-0.1995	179.65	380.673	False
6	4645	128.217	49.4579	2.5925	0.9226	0.72	-0.0979	201.844	870.357	False
7	2177	80.2406	35.2894	2.2738	0.8981	0.8959	-0.0663	304.003	624.629	False
8	3251	124.464	40.9538	3.0391	0.9443	0.639	0.028	423.426	66.5638	False
9	3239	125.767	33.4856	3.7559	0.9639	0.9114	-0.0886	481.731	573.277	False
10	3188	122.408	33.7355	3.6285	0.9613	0.8804	-0.0452	529.624	391.03	False
11	3013	170.645	29.6128	5.7626	0.9848	0.5248	-0.2083	594.958	912.812	False
12	3414	148.999	33.1799	4.4906	0.9749	0.8294	-0.0805	580.581	782.628	False
13	4647	139.214	45.6902	3.0469	0.9446	0.8623	-0.0076	723.033	247.267	False
14	1123	92.9757	18.8182	4.9407	0.9793	0.6997	-0.1287	817.288	724.433	False
15	3750	145.258	33.093	4.3894	0.9737	0.8965	-0.0485	952.204	400.276	True
16	1341	90.1135	19.5596	4.6071	0.9762	0.911	-0.0956	984.752	948.262	True
17								310.754	226.539	False
18								933.644	3.9103	True
19								368.378	576.748	False

Network per lacuna

lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px

1	7	218	587	843.269	34	24.802	7.8744
2	8	278	584	728.772	37	19.6965	8.655
3	14	376	696	1457.74	53	27.5045	7.6495
4	12	324	725	1290.64	52	24.8201	6.979
5	10	519	1176	3657.89	165	22.1691	7.5297
6	30	842	1719	4824.39	205	23.5336	7.5581
7	17	424	763	854.683	41	20.8459	7.2784
8	16	480	903	942.039	46	20.4791	7.6632
9	7	433	1012	2110.11	95	22.2117	8.5058
10	17	525	1324	3705.45	151	24.5394	7.655
11	27	912	1944	4228.87	235	17.9952	7.2482
12	22	653	1390	2435.61	123	19.8017	7.0574
13	20	650	1360	2447.77	125	19.5821	7.6042
14	12	433	976	3230.66	136	23.7549	7.3677
15	9	405	936	2447.3	99	24.7202	7.6472
16	6	240	595	278.936	13	21.4566	5.8744
17	20	699	1439	1649.89	82	20.1207	7.7675
18	3	112	250	494.132	25	19.7653	7.4358
19	5	168	250	286.581	17	16.8577	7.6535

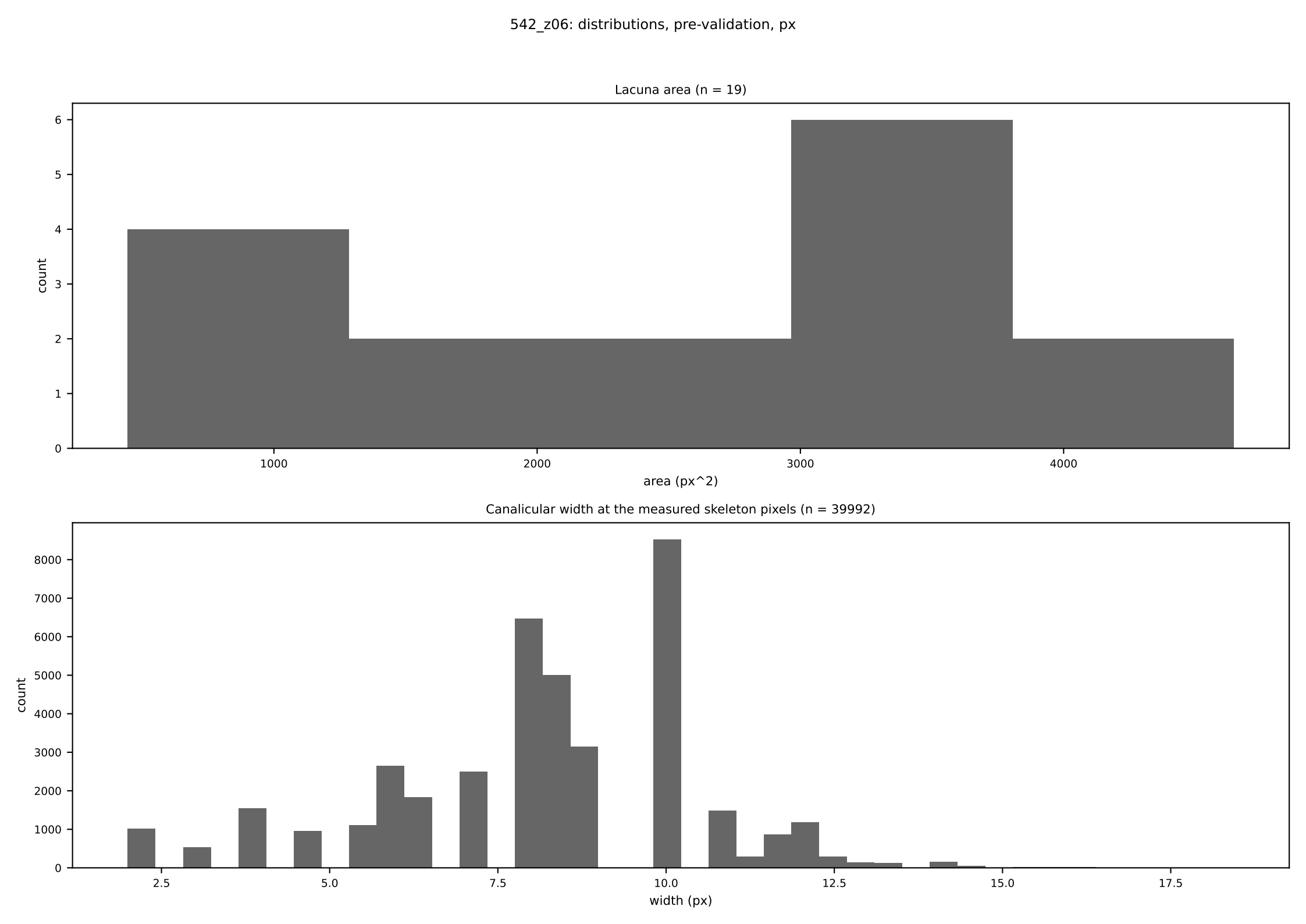
Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	215.232	5617	12694	0.6363	0.5983	0.0388107	0.0462423	3.2523
2	126.468	6226	14794	1	1	0.0446515	0.0394755	6.3257
3	225.309	8379	19327	0.9998	0.884	0.0448741	0.0360118	6.2137
4	151.782	6752	15321	1	1	0.0479858	0.0473207	7.9061
5	322.043	11663	28685	1	1	0.0444997	0.040997	3.1052
6	438.517	13901	31616	1	1	0.0605712	0.0543712	6.8412
7	211.924	8423	18846	1	1	0.0503384	0.040486	8.0217
8	359.191	12383	26961	1	0.9025	0.0387628	0.0334928	4.4545
9	302.451	10940	24775	1	1	0.0395795	0.0408476	2.3144
10	309.522	11337	28286	1	1	0.0463085	0.0468076	5.4923
11	468.274	15363	34096	1	1	0.0593634	0.0570155	5.7659
12	422.978	13165	30001	1	1	0.0496012	0.0463318	5.2012
13	379.848	13296	32108	1	1	0.0488869	0.042357	5.2653
14	217.924	8830	22986	1	1	0.0490374	0.0424606	5.5065
15	332.894	10505	23806	0.8602	0.7919	0.0385531	0.0393178	2.7036
16	205.776	6766	16357	0.7841	0.714	0.0354715	0.0363759	2.9158
17	387.948	13037	31120	1	1	0.0536166	0.0462404	5.1553
18	87.0061	2915	8653	0.5433	0.5282	0.038422	0.0288917	3.448
19	56.8701	3677	7091	1	1	0.0456894	0.035256	8.792

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	218	580	251.48	670.115	6	5	5
2	259	453	316.179	657.973	5	8	8
3	316	641	432.404	791.654	8	7	10
4	320	644	365.836	814.299	6	7	10
5	468	988	607.428	1357.56	11	17	15
6	839	1624	970.749	1977.11	23	20	22
7	362	558	479.919	866.411	8	13	12
8	436	711	539.375	1011.08	11	13	10
9	378	645	488.889	1135.48	12	13	12
10	525	1315	602.316	1518.33	11	13	16
11	843	1867	1047.02	2202.63	17	23	25

12	611	1219	734.5	1570.18	16	18	15
13	617	1262	736.227	1539.43	15	23	19
14	433	846	487.433	1106.16	9	12	12
15	389	799	460.404	1056.78	13	11	16
16	224	259	273.35	667.428	7	7	8
17	654	1073	796.441	1653.98	13	16	18
18	112	241	127.74	281.723	3	3	5
19	106	122	183.711	275.48	2	6	6


542_z06: distributions, pre-validation, px
Lacuna area (n = 19)
count
area (px^2)
Canalicular width at the measured skeleton pixels (n = 39992)
count
width (px)